

ASSIGNMENT 2



Topic:

Assignment 1: Customer segmentation using cluster analysis with the restaurant data

Module:

MGT7215

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Introduction

Managerial efforts to manage business interactions with customers by combining business processes and technologies that seek to understand a company's customers (Kim, 2003). The method of using data and statistical techniques in marketing analytics aims to better understand consumer behavior and guide marketing decisions. Businesses can increase consumer pleasure by improving their products, but the resources they have often aren't enough (Xu, 2009). Because of this, it is crucial to optimize customer happiness while working with limited resources to develop products (Violante, 2017). Four factors form the foundation of the primary conceptual perspective for customer relationship management. These four dimensions are customer identification, customer attraction, customer retention, and customer development (Ngai, 2009). Customer segmentation, which involves grouping customers based on shared traits or behaviors, is a key tactic in marketing analytics. An approach frequently used for customer segmentation is cluster analysis. It is carried out to Organize customers into groups based on shared traits and interests, and then create targeted marketing plans to satisfy the specific requirements of each group. In this study, the data exploration is carried out using variables by the means of tableau. The objective is to find out the customer preferences and to follow the ideal marketing campaign, hierarchical clustering method and k means clustering method is carried out to find out the cluster which has highest preference for the administrative people to open a new store.

Literature review

In this approach, (Wu, 2005) author used a bank's 2003 credit card data to do the research on the consumer segmentation model, Customers were categorized into 9 different consumption levels based on the results of the clustering study and was divided into A1, A2..., A9 which denote the levels from the lowest to the highest. Consumption fluctuation has six, and thus 54 groups of clients were obtained with various axes of features. Variety of customers are represented by a mode and level combination. In a study, conducted by (Khalili-Damghani, 2018) based on past consumer behaviors in distinct segmentations, the author used K-Means to forecast upcoming transactions. The hybrid feature selection for filtering and decision-making method is employed to help with this. To predict sales volume and business profit for a company, the current study can also be used in the insurance and telecommunications sectors. This approach is particularly good at forecasting profitable clients as well as spotting new customer behavior. In this study, Retail products are clustered based on market basket data using a data-driven methodology. K-means, SOM, and hierarchical clustering techniques are used to compare the results. They discovered that the lack of receipts with a bigger quantity of various products leads to a violation of the ideal behavior and the method's assumptions. The application employing actual data from a Czech drugstore company demonstrates that their method produces results that are comparable to the classification. In the Czech drugstore industry, this is a typical element of customer behavior (Holý, 2017).

Customer segmentation

According to (Sarigöllü, 2005) Customer segmentation refers to the process of dividing customers into groups based on their distinct needs, characteristics, or behaviors, which is commonly used in marketing research to enhance existing product marketing strategies and create new ones. This technique can provide a competitive advantage in a market with numerous service providers, as it allows businesses to understand the market's nature and the various target categories (Bojanic, 2007). There are two primary methods for categorizing clients for segmentation purposes: a priori segmentation, where grouping criteria are predetermined, and post hoc segmentation, which uses quantitative data analysis techniques, such as cluster analysis, to identify categories (Mauri, 2013). In the Restaurant industry, effective marketing strategies can be established by understanding the different target categories.

Cluster analysis

Data are grouped together into similar groupings through the process of clustering. A cluster is a collection of data that are most unlike to the data in other clusters and have the closest resemblance and proximity to one another within that cluster (Khajvand, 2011). Previous studies have widely used the K-means clustering

algorithm for segmenting purposes. This algorithm has been applied in different areas such as data mining, statistical data analysis, and other business-related applications (Cheng, 2009). Due to its computational simplicity, K-means clustering is famously used.

Methodology

Data exploration

Firstly, import the “Restaurant_Data” through tableau platform for data exploration. Compare the variable displayed on the left to explore the customer preference, characters or behavior. In this case, the variable quality of service is presented using zip code to indicate the which of the states has good quality of rating according to customer preference. Another variable as prioritize hygiene has been visualized.

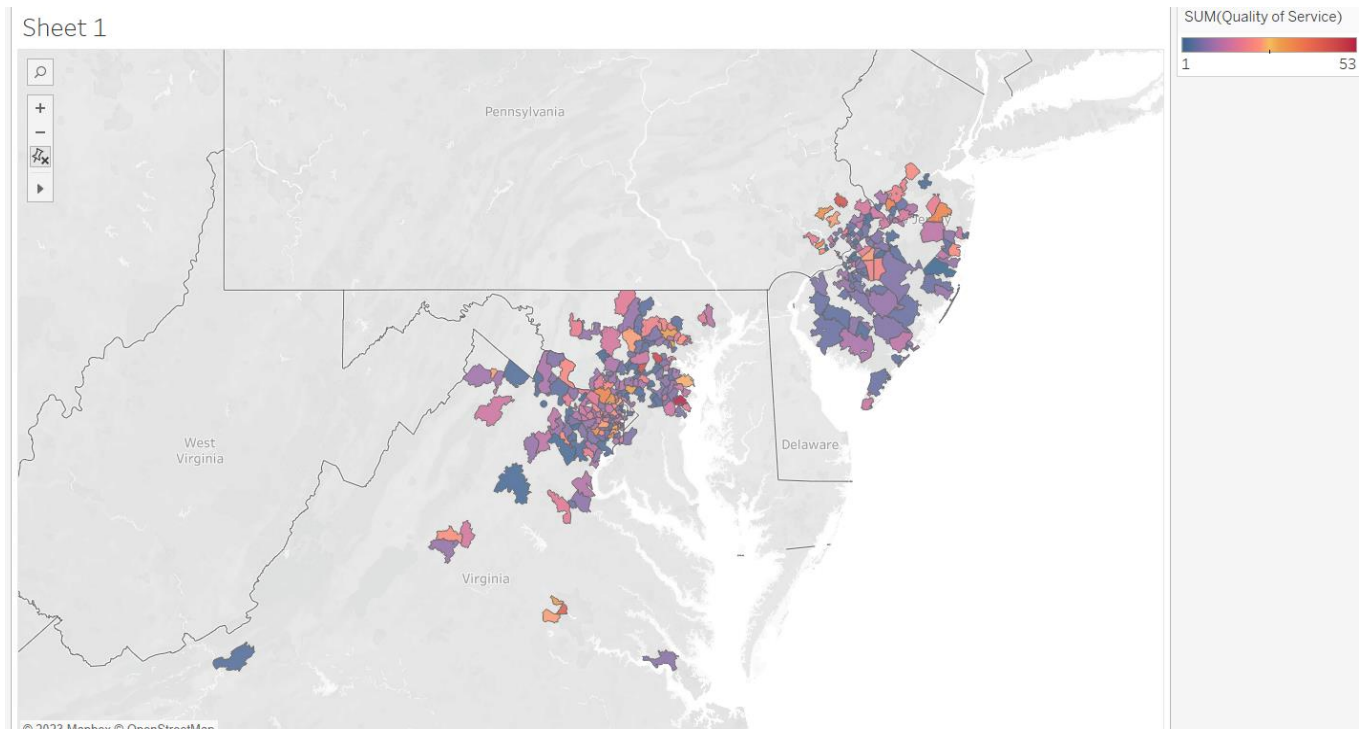


Figure 1 quality of service in each zip codes

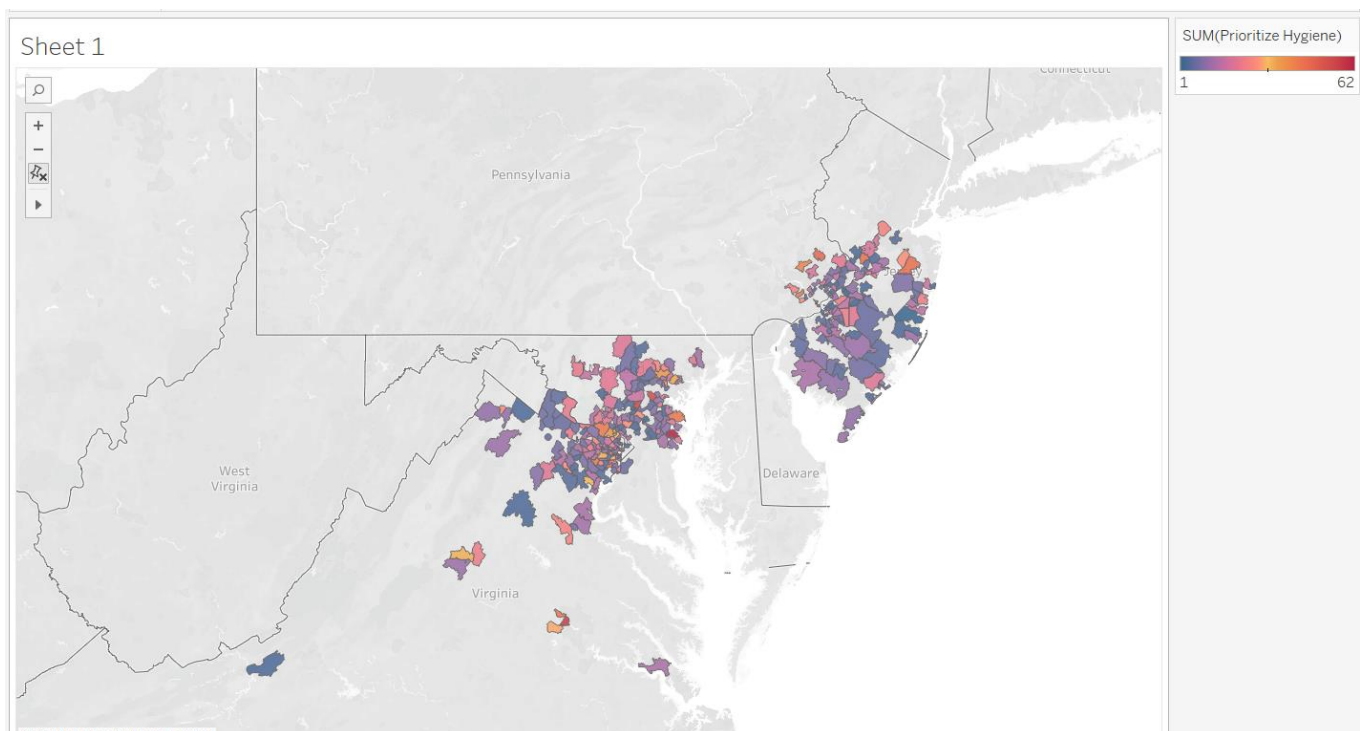


Figure 2 restaurants who prioritize hygiene

Hierarchical clustering model

Initially, the packages must be loaded according to the necessary requirement, and the seed must be set. The data must be imported through the read excel function. Clustering customers of a restaurant is done based on eight variables related to their behavior and preferences. The first step involves creating a distance matrix using the dist function on the variables. This function calculates the distance between each pair of customers based on their values for the eight variables. The eight variables used for clustering are average order size, average order frequency, reasonable pricing, quality of service, staff behavior, prioritizing hygiene, menu design, and beverages. Next, hierarchical clustering models are created using the hclust function with different linkage techniques such as single, average, centroid, and complete. In contrast to complete linkage, which generates clusters based on the longest distance between any two points in each cluster, single linkage bases its clustering on the shortest distance between any two points in each cluster. The median and centroid distances between the points in each cluster serve as the foundation for the average and centroid linkage approaches, respectively. An elbow plot is then created to determine the optimal number of clusters for the final model. The plot's "elbow" point, where the rate of decline shows the ideal number of clusters. Within cluster variation calculates the overall separation between each location and the cluster center. A dendrogram is produced to visualize the outcomes of the hierarchical clustering, which is a tree like structure that represents each branch with a cluster. The scale function is used to ensure there is no greater impact on the clustering result due to difference in scales.

K-means clustering model

The kmeans function is used to perform K-means clustering on the scaled variables with three clusters. This method updates the centroids based on the updated cluster assignments, assigning each location to the cluster where the centroid is closest to it. To evaluate the effectiveness of the clustering process, the within cluster variation is computed for each cluster. This metric measures the total distance between each data point and the center of its corresponding cluster. A lower within cluster variation value indicates that the cluster is more compact and the data points are closer to the center of the cluster. To compute the total within cluster variation for all clusters, the function tot.withinss is used. We create an additional variable called segmentation that holds the cluster assignments to incorporate the cluster assignments into the original data. The segmentation variable and the original data are then combined using the cbind() method, and the outcome is then assigned to a brand-new variable named segment results. Lastly, we export the segment results data frame to a CSV file using the write.csv() function. The resulting clusters are then added to the original data, and the data is exported to a CSV file for further analysis.

Visualization of the customer segmentation

Browse to the location where you saved the CSV file. Click 'Open' to import the file. In the tab, you can see the data columns and rows imported from the CSV file. To create a bar chart with prioritize hygiene variable, drag the variable to the 'Rows' and 'Columns' sections respectively. To differentiate the data points based on their cluster, drag the 'segmentation' column to the 'Color' section. The CSV file is exported and imported into tableau for the visualization. Using the segmented data sets we will be able to visualize which cluster highly prefers the specific variables from the dataset.

Sheet 1

	Segmentation		
	1	2	3
Avg. Avg Order Freq	1.51	1.60	1.69
Avg. Avg Order Size	36.51	36.20	34.87
Avg. Beverages	5.55	3.69	4.97
Avg. Menu Design	3.75	2.96	5.30
Avg. Prioritize Hygie..	4.66	3.17	4.77
Avg. Quality of Servi..	2.40	3.71	4.31
Avg. Reasonable Pri..	1.95	4.00	3.47
Avg. Segmentation	1.00	2.00	3.00
Avg. Staff behavior	2.86	3.35	5.31

Figure 3 customer segmentation

For example, the variable reasonable pricing in cluster 2 is high, this explains that there is a fair amount of price given in the restaurants that is preferred within the customers.

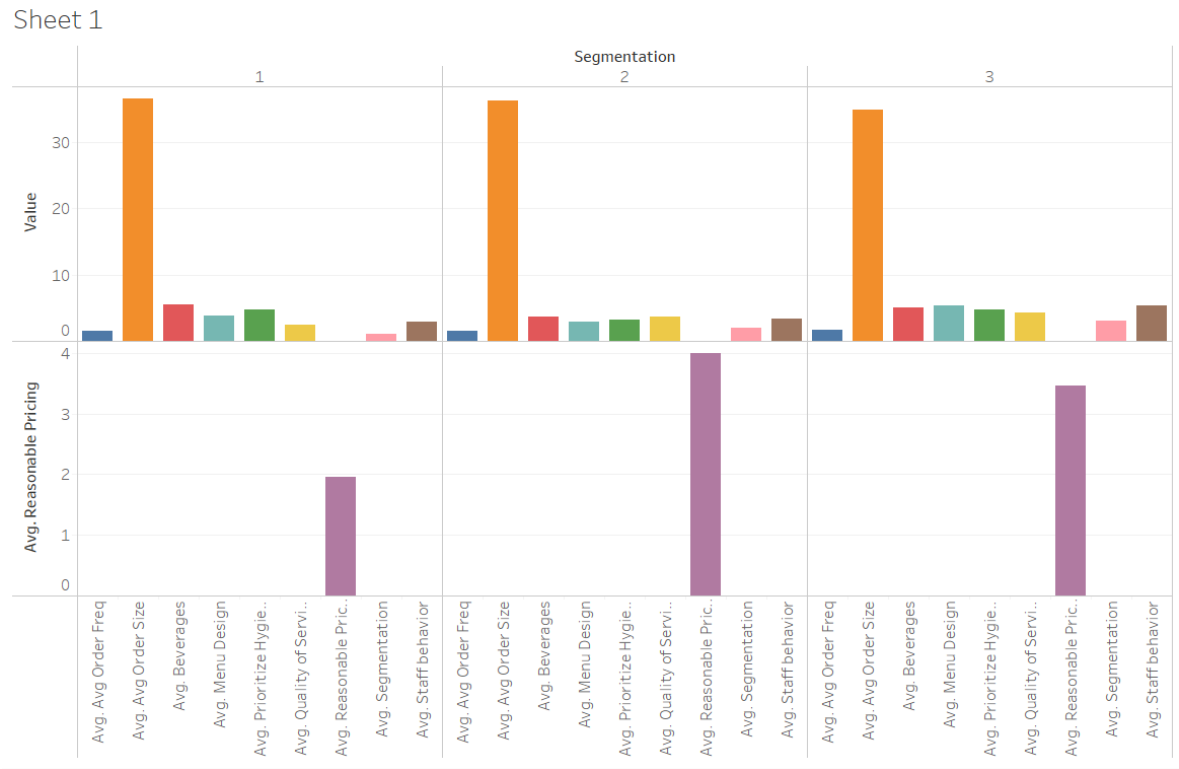


Figure 4 Reasonable pricing

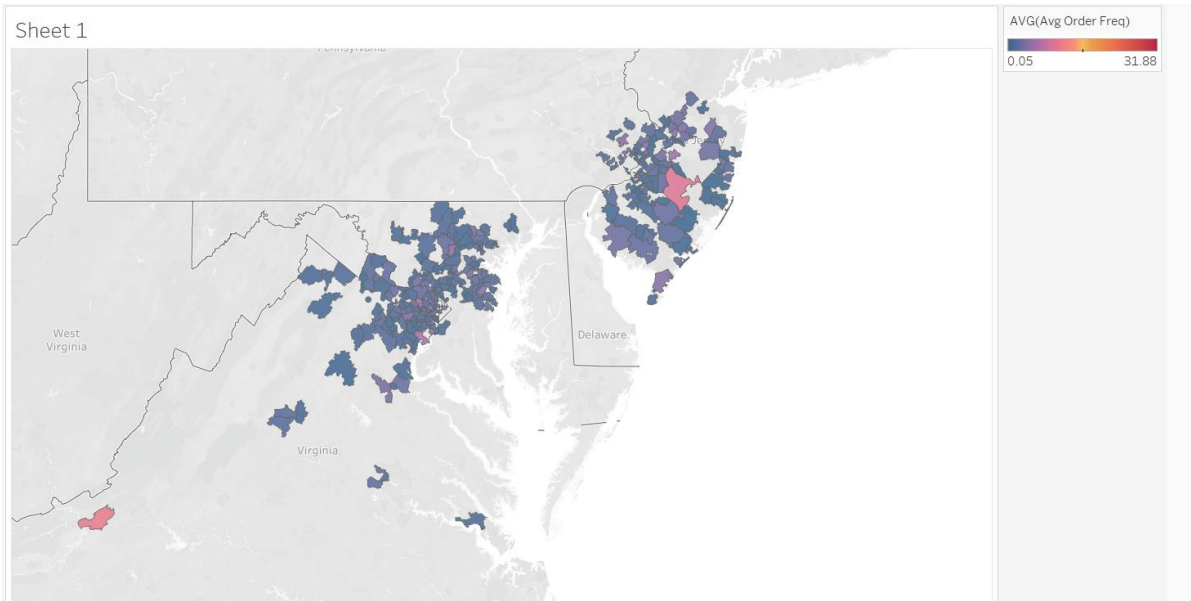


Figure 5 Average order frequency

Results and discussion

The elbow plot revealed that three clusters are the ideal amount for this dataset. Three primary clusters could be seen on the dendrogram. Three clusters were created using K-means clustering, and for each cluster, the within cluster variation was calculated. Several outcomes were produced by the hierarchical clustering models created by using various linkage on the distance matrix of the scaled variables. The dendrogram produced by the single linkage method, which connects all the data points in a long chain, may not be the most useful visual representation of the data. The dendrogram produced by the average linkage method has clusters that were dispersed equally, which might be a sign that the dataset lacks distinct group. The elbow plot demonstrated that 3 clusters are the optimum number for this dataset, and the dendrogram also demonstrated this.

The scaled variables underwent K-means clustering with three clusters. Both k-means clustering and hierarchical clustering point to the existence of three separate customer segments. The dendrogram demonstrates how well-separated the three groups are from one another, with one cluster being much farther away from the other two. This is supported by the three distinct clusters found in the k-means clustering data. The three-cluster approach appears to be appropriate based on the within cluster sum of square (WCSS) variation values.

Cluster means:				
	segment.avg_order_size	segment.avg_order_freq	segment.Reasonable_Pricing	segment.Quality_of_Service
1	83.43907	0.6882894	3.109890	3.329670
2	391.41667	0.2799423	3.000000	3.666667
3	29.89500	1.6989541	3.227373	3.537528
	segment.Staff_behavior	segment.Prioritize_Hygiene	segment.Menu_Design	segment.Beverages
1	3.769231	3.736264	3.791209	4.670330
2	4.333333	3.333333	4.666667	5.000000
3	3.858720	4.183223	3.981236	4.657837

Figure 6 segmented clusters

The K-means clustering findings indicate that the dataset is divided into three different regions. Customers in Cluster 2 are rated highest for all quality-related characteristics, with the highest average order size, staff behavior, menu design, beverages and Quality of service. This cluster may be seen of as the most valuable one for the restaurant, therefore it should concentrate on keeping and growing this group of customers. All variables in cluster 2 have high ratings, indicating a subset of customers that are very satisfied. Cluster 3 had the lowest scores across all factors, suggesting that this group might have a particular preference for the restaurant's food.

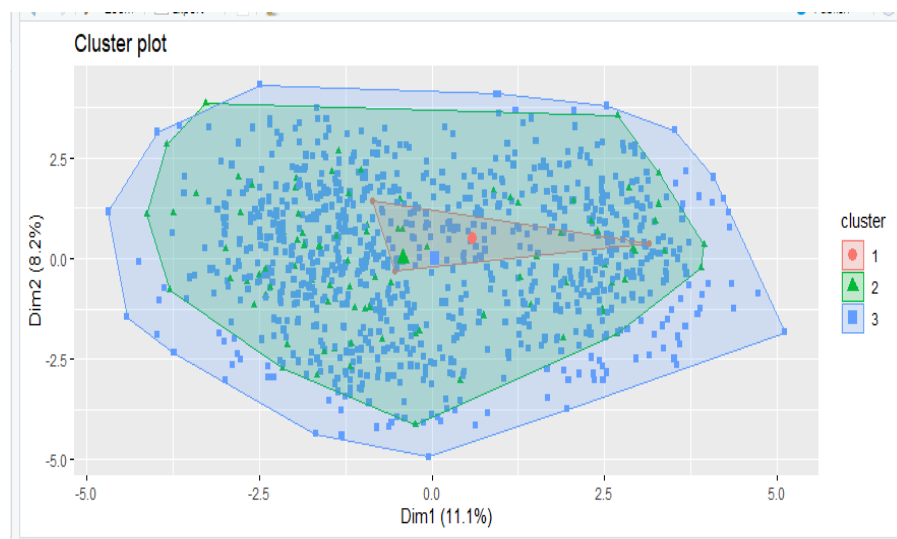


Figure 7 cluster plot

The clients who are the most satisfied with every aspect of their eating experience are represented by the green cluster at the top. Customers with a moderate level of satisfaction with the majority of the dining experience's features are represented by the red cluster in the center. The customers who are least satisfied with the majority of the dining experience's features are represented by the blue cluster at the bottom. Ultimately, the segmentation analysis offers insightful information about the restaurant's customers that can be directed to create focused marketing campaigns and enhance the customer experience.

Conclusion

This method helps restaurants better understand their consumer segments and offers insights into the dining experiences of customers. The restaurant can discover areas that need development and create strategies to raise client satisfaction by studying the data connected to the eating experience. Since the findings of clustering constitute the foundation for important and frequently irreversible marketing decisions, it is important to debate how to advance theoretical and practical CA applications in the future (Yankelovich,

2006). Customers' requirements, though, are not always constant and can result in the market changing dramatically. According to this, segment instability can be both a gift and a curse (Blocker, 2007). The restaurant can also create customized marketing campaigns and modify its services to fit the demands of each category by grouping its consumers into several segments. In order to position itself differently for each market, a hotel restaurant may be able to recognize and describe three distinct market segments. Additionally, the method can be used to reveal overlooked but useful market groups (Jurowski, 2000). It depends on the restaurants to prioritize the marketing strategy according to the segments and lookout for the profitable end.

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Appendix

```
#Load packages
```

```
install.packages("cluster")
```

```
install.packages("factoextra")
```

```
install.packages("readxl")
```

```
install.packages("tidyverse")
```

```
install.packages("tidyr")
```

```
install.packages("corrplot")
```

```
library(factoextra)
```

```
library(readxl)
```

```
library(cluster)
```

```
library(tidyverse)
```

```
library(tidyr)
```

```
library(dplyr)
```

```
library(ggplot2)
```

```
library(corrplot)
```

```
setwd("D:/marketing analytics")
```

```
## Set Seed
```

```
set.seed(1234)
```

```
# Import Data from excel
```

```
segment <- read_xlsx(file.choose())
```

```
#scaling Distance
```

```
segment1.dist = dist(segment)
```

```
segment1.dist
```

```
###hierarchical clustering with linkage techniques###
```

```
hierarchial.cluster1 <- hclust(dist(scale(cbind(segment$avg_order_size, segment$avg_order_freq,  
segment$Reasonable_Pricing,
```

```
segment$Quality_of_Service,segment$Staff_behavior,
```

```
segment$Prioritize_Hygiene,segment$Menu_Design,segment$Beverages ))), method =
```

```
"single")
```

```
hierarchial.cluster2 <- hclust(dist(scale(cbind(segment$avg_order_size, segment$avg_order_freq,  
segment$Reasonable_Pricing,
```

```
segment$Quality_of_Service,segment$Staff_behavior,
```

```
segment$Prioritize_Hygiene,segment$Menu_Design,segment$Beverages ))), method =
```

```
"average")
```

```
hierarchial.cluster3 <- hclust(dist(scale(cbind(segment$avg_order_size, segment$avg_order_freq,  
segment$Reasonable_Pricing,
```

```
segment$Quality_of_Service,segment$Staff_behavior,
```

```
segment$Prioritize_Hygiene,segment$Menu_Design,segment$Beverages ))), method =
```

```
"centroid")
```

```
hierarchial.cluster4 <- hclust(dist(scale(cbind(segment$avg_order_size, segment$avg_order_freq,  
segment$Reasonable_Pricing,
```

```
segment$Quality_of_Service,segment$Staff_behavior,
```

```
segment$Prioritize_Hygiene,segment$Menu_Design,segment$Beverages ))), method =
```

```
"complete")
```

```
#elbow Plot
```

```
x <- c(1:8)
```

```
sort_height <- sort(hierarchial.cluster4$height,decreasing=TRUE)
```

```
y <- sort_height[1:8]
```

```
plot(x,y); lines(x,y,col="red")
```

```

#plot dendrogram

plot(hierarchial.cluster4)

rect.hclust(hierarchial.cluster4, k = 3 )


#scaling for K means

segment2 <- select(segment,avg_order_size, avg_order_freq,Reasonable_Pricing,
                    Quality_of_Service,Staff_behavior,
                    Prioritize_Hygiene,Menu_Design,Beverages)


segment3 <- scale(segment2)


#k-mean cluster analysis

segment_kmeans <- kmeans(x = data.frame(segment$avg_order_size, segment$avg_order_freq,
segment$Reasonable_Pricing,
                    segment$Quality_of_Service,segment$Staff_behavior,
                    segment$Prioritize_Hygiene,segment$Menu_Design,segment$Beverages), 3)

segment_kmeans


kmeans_output <- kmeans(segment_kmeans$cluster, centers = 3)

fviz_cluster(kmeans_output,data = segment, geom = "point")


#within cluster variation#

segment_kmeans$withinss


#within cluster variations for different cluster#

segment_kmeans$tot.withinss


# Adding segment number to original data for excel file#

segmentation= km.out$cluster

segment_results <- cbind(segment, segmentation)


# Export data#

```

```
write.csv(segment_results, file = "D:/marketing analytics/output.csv", row.names = FALSE)
```